

Bacterial Biomarkers

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Introduction

Bacterial surface markers represent a diverse array of molecular structures that are uniquely expressed on the exterior of bacterial cells and play crucial roles in their identification, pathogenesis, and interaction with the host environment. Unlike genetic markers that may be shared across microorganisms through horizontal gene transfer, surface markers often exhibit high species-specificity due to their evolutionary adaptation to particular environmental niches and host interactions.

In the context of rising antimicrobial resistance (AMR) and the need for precise therapeutics, accurately identifying bacterial pathogens using these specific surface markers has become increasingly critical in clinical microbiology.¹

Focus Area

- Markers specific to the surface of bacteria and not overlapping with other pathogens / Readily available or expressable markers only in bacteria
- Help healthcare professionals with confirmation of a bacterial infection and subsequent gram classification for early diagnosis and treatment.
- SERS based detection .

¹ Kotoulas NK, Tan S, Nodwell JR, Goh MC. Differentiating bacteria by their unique surface interactions. PLoS One. 2025 Jun 30;20(6):e0327489. doi: 10.1371/journal.pone.0327489. PMID: 40587576; PMCID: PMC12208452.

Current Limitations :

Traditional sepsis biomarkers like (PCT, CRP, IL-6, Presepsin etc) are faced with challenges because they detect the host's inflammatory response and not the bacterial pathogen itself. These markers may also get elevated in viral infections (COVID-19, influenza), fungal sepsis (Candida, Aspergillus), parasitic diseases (malaria), autoimmune conditions (lupus, rheumatoid arthritis), malignancy, burns, surgery, and drug reactions .^{2 3}

This non-specificity creates a diagnostic dilemma. Example : elevated PCT/CRP could indicate a bacterial sepsis requiring antibiotics, or it could be just a viral pneumonia, autoimmune flare up, localised infection or post-surgical inflammation where antibiotics can prove to be harmful.

Clinicians are forced to treat empirically fueling the antimicrobial resistance crisis affection innumerable patients globally. On the other hand our Bacterial exclusive biomarkers can try to overcome this uncertainty by detecting only bacterial pathogen, providing definitive evidence that bacteria—not viruses, fungi, or inflammation—are present along with subsequent gram classification enabling precision antimicrobial therapy will help in improved clinical outcomes and reduced anti-microbial resistance.

Traditional biomarkers ask "Is the immune system fighting?" Our biomarkers can answer "Are bacteria actually present?" That's the difference between reactive guesswork and precision medicine.

<u>Traditional Approach</u>	<u>Bacterial-Specific Approach</u>
Host response detection	Direct bacterial detection
Delayed immune activation	Immediate bacterial identification
Non-specific inflammation	Bacterial specific marker
Cannot guide antibiotics	Enables targeted therapy
Incremental improvement	Paradigm shift

Following are the Bacterial Exclusive surface Biomarkers which can help in diagnosis of a bacterial infection along with the determination of the pathogens Gram category (Positive / Negative) with consideration of SERS based feasibility.

² Evans L, Rhodes A, Alhazzani W, Antonelli M, Coopersmith CM, French C, Machado FR, Mcintyre L, Ostermann M, Prescott HC, Schorr C, Simpson S, Wiersinga WJ, Alshamsi F, Angus DC, Arabi Y, Azevedo L, Beale R, Beilman G, Belley-Cote E, Burry L, Cecconi M, Centofanti J, Coz Yataco A, De Waele J, Dellinger RP, Doi K, Du B, Estenssoro E, Ferrer R, Gomersall C, Hodgson C, Hylander Møller M, Iwashyna T, Jacob S, Kleinpell R, Klompas M, Koh Y, Kumar A, Kwizera A, Lobo S, Masur H, McGloughlin S, Mehta S, Mehta Y, Mer M, Nunnally M, Oczkowski S, Osborn T, Papathanassoglou E, Perner A, Puskarich M, Roberts J, Schweickert W, Seckel M, Sevransky J, Sprung CL, Welte T, Zimmerman J, Levy M. Surviving Sepsis Campaign: International Guidelines for Management of Sepsis and Septic Shock 2021. Crit Care Med. 2021 Nov 1;49(11):e1063-e1143. doi: 10.1097/CCM.0000000000005337. PMID: 34605781.

³ National Institute for Health and Care Excellence. Sepsis: recognition, diagnosis and early management. NICE guideline [NG51]. September 2017.

1.Universal Bacterial Detection

Peptidoglycan (PGN) :

- ★ N-acetylmuramic acid
- ★ N-acetylglucosamine
- ★ D-amino acids

Exclusivity:

- Present in ALL bacteria, **absent** in viruses/fungi/parasites .
- Direct bacterial confirmation
- Unique to bacterial peptidoglycan only.^{4 5 6}

2. Gram classification

2.a Gram-Positive–Specific Markers

- ★ Lipoteichoic Acid (LTA)
- ★ Wall Teichoic Acid (WTA)

Uniqueness: Only in Gram-positive bacteria , surface accessibility .^{7 8 9}

⁴ Hassan M, Zhao Y, Zughaier SM. Recent Advances in Bacterial Detection Using Surface-Enhanced Raman Scattering. Biosensors (Basel). 2024 Aug 1;14(8):375. doi: 10.3390/bios14080375. PMID: 39194603; PMCID: PMC11352333.

⁵ Yu Yang, Chuan Zeng, Jing Huang, Mengfan Wang, Wei Qi, Haixia Wang, Zhimin He, Specific and quantitative detection of bacteria based on surface cell imprinted SERS mapping platform, Biosensors and Bioelectronics, Volume 215, 2022, 114524, ISSN 0956-5663, <https://doi.org/10.1016/j.bios.2022.114524>.

⁶ Vollmer W, Blanot D, de Pedro MA. Peptidoglycan structure and architecture. FEMS Microbiol Rev. 2008 Mar;32(2):149-67. doi: 10.1111/j.1574-6976.2007.00094.x. Epub 2008 Jan 8. PMID: 18194336.

⁷ Schneewind O, Missiakas D. Lipoteichoic acids, phosphate-containing polymers in the envelope of gram-positive bacteria. J Bacteriol. 2014 Mar;196(6):1133-42. doi: 10.1128/JB.01155-13. Epub 2014 Jan 10. PMID: 24415723; PMCID: PMC3957714.

⁸ Guoqing Xia, Thomas Kohler, Andreas Peschel, The wall teichoic acid and lipoteichoic acid polymers of Staphylococcus aureus, International Journal of Medical Microbiology, Volume 300, Issues 2–3, 2010, Pages 148-154, ISSN 1438-4221, <https://doi.org/10.1016/j.ijmm.2009.10.001>.

⁹ Stephen KE, Homrighausen D, DePalma G, Nakatsu CH, Irudayaraj J. Surface enhanced Raman spectroscopy (SERS) for the discrimination of Arthrobacter strains based on variations in cell surface composition. Analyst. 2012 Sep 21;137(18):4280-6. doi: 10.1039/c2an35578g. Epub 2012 Jul 30. PMID: 22842541; PMCID: PMC3419301.

2b. Gram Negative specific marker

★ Lipopolysaccharides (LPS) : Lipid A , O antigen

Uniqueness:

- Only in Gram-negative bacteria, absent in all other organisms
- Strong enhancement, endotoxin correlation .^{10 11 12}

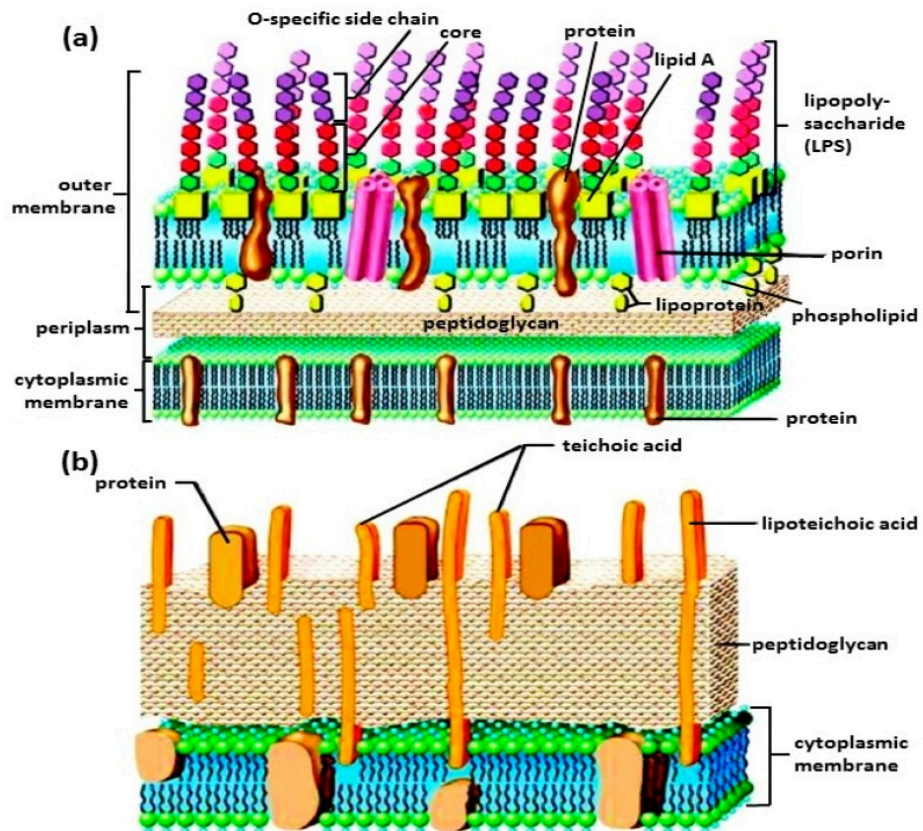


Fig: Representation of bacterial cell wall a) Gram -ve b) Gram +ve

¹⁰ Cicchinelli S, Pignataro G, Gemma S, Piccioni A, Picozzi D, Ojetti V, Franceschi F, Candelli M. PAMPs and DAMPs in Sepsis: A Review of Their Molecular Features and Potential Clinical Implications. *Int J Mol Sci.* 2024 Jan 12;25(2):962. doi: 10.3390/ijms25020962. PMID: 38256033; PMCID: PMC10815927.

¹¹ Ji Z, Zhang C, Ye Y, Ji J, Dong H, Forsberg E, Cheng X, He S. Magnetically Enhanced Liquid SERS for Ultrasensitive Analysis of Bacterial and SARS-CoV-2 Biomarkers. *Front Bioeng Biotechnol.* 2021 Sep 29;9:735711. doi: 10.3389/fbioe.2021.735711. PMID: 34660557; PMCID: PMC8511622.

¹² Rietschel ET, Kirikae T, Schade FU, Mamat U, Schmidt G, Loppnow H, Ulmer AJ, Zähringer U, Seydel U, Di Padova F, et al. Bacterial endotoxin: molecular relationships of structure to activity and function. *FASEB J.* 1994 Feb;8(2):217-25. doi: 10.1096/fasebj.8.2.8119492. PMID: 8119492.

3. Other adjunct / specific Markers :

Porins (OmpA/OmpF/OmpC)

- for species specific¹³⁻¹⁴

Calcium Dipicolinate (CaDPA) :

- Found in Bacterial spores only .¹⁵

Surface Proteins

- Pathogen identification (Protein A, PilA.)¹⁶

Braun's Lipoprotein

- abundant protein in *E. coli*; Enterobacteriaceae.¹⁷

Pentraxin 3 (PTX3)

- protein with structural similarity to CRP, which may be produced primarily by inflammatory cells rather than the liver. Like CRP, elevated levels of PTX3 have been shown to correlate with the severity of sepsis¹⁸

Conclusion: Based on comprehensive analysis of current literature, suggestive priority markers for the SERS-based detection of bacterial sepsis.

- **For universal bacterial detection**, components of the peptidoglycan layer (particularly N-acetylmuramic acid) can be good targets as they are present in all bacteria but absent in other pathogens.
- For **Gram-negative bacteria**, lipopolysaccharides (LPS) represent the most characteristic surface markers,
- For **Gram-positive bacteria**, focusing on teichoic acids as primary detection targets due to their abundance and highly specific Raman signatures can be beneficial .

These markers can be supplemented with bacterial-specific components to enhance detection sensitivity in complex cases where signals might be obscured.

¹³ Hassan M, Zhao Y, Zughaier SM. Recent Advances in Bacterial Detection Using Surface-Enhanced Raman Scattering. Biosensors (Basel). 2024 Aug 1;14(8):375. doi: 10.3390/bios14080375. PMID: 39194603; PMCID: PMC11352333

¹⁴ Stephen KE, Homrighausen D, DePalma G, Nakatsu CH, Irudayaraj J. Surface enhanced Raman spectroscopy (SERS) for the discrimination of Arthrobacter strains based on variations in cell surface composition. Analyst. 2012 Sep 21;137(18):4280-6. doi: 10.1039/c2an35578g. Epub 2012 Jul 30. PMID: 22842541; PMCID: PMC3419301.

¹⁵ Segervald J, Malyshev D, Öberg R, Zäll E, Jia X, Wågberg T, Andersson M. Ultra-Sensitive Detection of Bacterial Spores via SERS. ACS Sens. 2025 Feb 28;10(2):1237-1248. doi: 10.1021/acssensors.4c03151. Epub 2025 Jan 23. PMID: 39847439; PMCID: PMC11877637.

¹⁶ Identification of Bacterial Surface Antigens by Screening Peptide Phage Libraries Using Whole Bacteria Cell-Purified Antisera JOURNAL=Frontiers in Microbiology VOLUME=Volume 8 - 2017

¹⁷ Schultz, J., Jamil, T., Sengupta, P. et al. Genomic insights into novel extremotolerant bacteria isolated from the NASA Phoenix mission spacecraft assembly cleanrooms. Microbiome 13, 117 (2025). <https://doi.org/10.1186/s40168-025-02082-1>

¹⁸ Zhang Y, Li X, Zhang X, Wang T, Zhang X. Progress in the study of pentraxin-3(PTX-3) as a biomarker for sepsis. Front Med (Lausanne). 2024 Jul 3;11:1398024. doi: 10.3389/fmed.2024.1398024. PMID: 39021820; PMCID: PMC11251883.